2024年吉林省高中生奥林匹克化学竞赛选拔赛参考答案

一、选择题。（每题2分，共15题，共30分）

题号	1	2	3	4	5	6	7	8
答案	D	D	C	B	A	D	B	C
题号	9	10	11	12	13	14	15	
答案	A	B	D	A	B	D	C	

二、选择题。（每题2分，部分选对得1分，共5题，共10分）

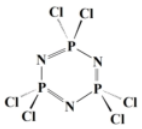
题号	16	17	18	19	20
答案	BC	D	BC	AC	BD

三、非选择题（共5题，每空1分，共60分）

21. (1) ① $4NaH+B(OCH_3)_3 \xlongequal{240^oC} NaBH_4+3CH_3ONa$ ②AB ③ $Na^+\left[H:\overset{H}{\underset{H}{B}}:H\right]^-$

(2) ① 2 3 圆底烧瓶 ② 蒸馏

(3) ① B D ② $5I^-+IO_3^-+6H^+=3I_2+3H_2O$ ③ 76 偏低

22. (1)$3s^23p^3$ (2)P_4S_{10} sp^3 (3)$2FeP_2+12FeS_2=P_4S_{10}+14FeS$ (4) 

(5)6 10:3 (6) (1,1,3/4) $\dfrac{290.5}{N_A m^2 n\times10^{-30}}$ (7)7/32 $3d^6$ (8)17:15 4

23. (1)$2MoS_2+7O_2 \xlongequal{高温} 2MoO_3+4SO_2$ (2) $MoO_2+2OH^-+ClO^-=MoO_4^{2-}+Cl^-+H_2O$

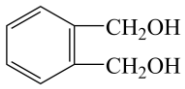
(3)SiO_2 (4)94.5% (5)冷却结晶 (6)焙烧产物中还会存在 MoO_2，稀硝酸具有强氧化性，可使 Mo 元素被进一步充分氧化，提高利用率 (7)浓氨水 (8)取最后一次洗涤液于试管中，加入硝酸酸化的硝酸银溶液，若没有白色沉淀生成，说明钼酸洗涤干净

(9) KI 溶液 当最后半滴指示剂加入时，锥形瓶内产生亮黄色沉淀，且 30 秒内不消失 $\dfrac{206cV}{m}\%$ (10)12 $(NH_4)_2MoO_4+21HNO_3+H_3PO_4=(NH_4)_3[PMo_{12}O_{40}]+21NH_4NO_3+12H_2O$

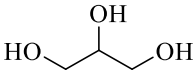
24. (1)BCD (2)$2N_2+6H_2O(g) \xlongequal{Fe_2O_3} 4NH_3+3O_2$ (3)$N_2+6e^-+6C_2H_5OH=2NH_3+6C_2H_5O^-$

(4) II T_2 反应 I 的 $\Delta H<0$，温度升高，平衡逆移，$c(N_2O_2)$减小，浓度降低的影响大于温度对反应 II 速率的影响 (5) b $\dfrac{2}{P_0}$ 66.7% (6) −447.3

25. (1) $CH_3CH=CH_2 + Cl_2 \xrightarrow{500^oC} CH_2ClCH=CH_2 + HCl$ 取代反应

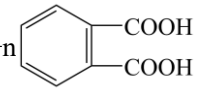
(2) 

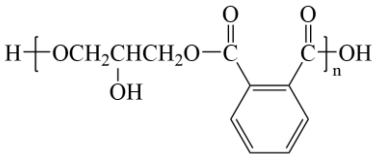
8

(3) $CH_2ClCHClCH_2Cl + 3NaOH \xrightarrow[\triangle]{H_2O}$ 

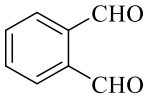
$+ 3NaCl$

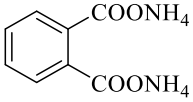
(4) 碳氯键，碳碳双键　　1,2,3-丙三醇或丙三醇或甘油

(5) $nCH_2OHCHOHCH_2OH+n$ 

$\rightarrow$ 

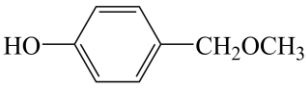
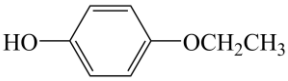
$+(2n-1)H_2O$

(6) 

$+ 4Ag(NH_3)_2OH \xrightarrow{\triangle}$ 

$+ 6NH_3 + 4Ag\downarrow + 2H_2O$

(7) HCHO、HCOOH　　(8) cd

(9) 16 

、